

Project CASCADE

A Contagion-Aware, Calibrated Decision System for

Event-Driven Traffic Congestion in Bengaluru

Flipkart × Bengaluru Traffic Police — Gridlock Hackathon 2.0 (Round 2)

The Hook. Most teams will hand the Bengaluru Traffic Police another *heatmap of where incidents happened*. CASCADE answers the question the police actually ask at 07:30 every morning: **“Given what is about to break, where do I put my ten tow-trucks and my officers *right now*, and how confident can I be?”** We do not just *predict* congestion — we *decide* the manpower, barricading, and diversion response, and we attach a **mathematical coverage guarantee** to every recommendation. The gap between a passive heatmap and a calibrated, contagion-aware dispatch plan *is* our project.

1. The Genesis: Ideation and Why This Approach

1.1 The Problem Statement and the Trap Most Solutions Fall Into

The ASTRAM dataset is an **incident-and-response log**, not a flow-sensor feed: 8,173 events over ~ 150 days (Nov 2023–Apr 2024), each carrying a cause, a location, a road-closure flag, a priority, and — crucially — timestamps that tell us *how long the road stayed blocked*. The obvious solution, and the one we expect from a large fraction of the field, is to cluster the points and render a hotspot map. We rejected that immediately for three reasons:

- **A heatmap is a description, not a decision.** The police objective is the allocation of scarce, physical resources (a documented constraint of ~ 10 tow-trucks across 150+ hotspots). A picture of the past does not place a tow-truck.
- **The labels are censored and messy.** Roughly 12.3% of incidents are still “active” (right-censored — their true duration is unknown), and $\sim 5,000$ “closed” rows carry no closure timestamp. Naïve regression on duration is statistically invalid here.
- **Incidents are contagious.** A breakdown at one junction spawns secondary incidents downstream. A model that treats events as independent ignores the very mechanism that turns a single stall into gridlock.

1.2 Why We Stand Out: From “Another Predictor” to a Decision System

The hackathon will produce many accurate-looking predictors that judges cannot act on. CASCADE is engineered around **four differentiators that competitors structurally lack**:

- **Survival under censoring, not regression.** We treat clearance time as a *survival* problem (DeepHit), so the 12.3% of unfinished incidents *inform* the model instead of being silently dropped — a correctness property, not a cosmetic one.
- **Contagion is a first-class citizen.** A self-exciting Hawkes intensity head forecasts *where and when the next incident will fire*, enabling *pre-positioning* rather than reaction. This is a capability no baseline possesses.
- **Provable trust.** Every duration estimate is wrapped in a **conformal coverage guarantee** (“ $\geq 90\%$ of incidents clear within X minutes” — and we empirically hit 90.7%). Judges and field officers can trust a number with a guarantee attached to it.
- **Decision-focused optimisation.** An OR-Tools allocator and a reinforcement-learning dispatcher convert calibrated predictions into officer coordinates, barricade lists, and diversion routes — with an equity constraint that breaks the documented predictive-enforcement feedback loop.

1.3 The Design Philosophy

We adopted a **tiered build**: Tier-0 is the smallest end-to-end slice that beats a strong baseline and demos completely; every later tier is a strictly additive upgrade that never breaks a working demo. “The system is the wow, not the kernel.” This let us de-risk delivery while still reaching an ambitious final architecture.

2. High-Level Overview: What CASCADE Computes

CASCADE ingests the raw incident log and continuously produces, per road-network junction, a bundle of decision-grade quantities:

- **De-biased hotspot surface.** A “naive” incident-count map versus a “corrected” congestion-impact map. The corrected surface weights each incident by severity, predicted duration, closure probability, and contagion, separating *exposure* (where tickets were logged) from *impact* (where the problem truly concentrates).
- **Calibrated clearance forecast.** For each junction: the median (P50) clearance time and a conformally-calibrated P90 upper bound (the coverage-guaranteed “worst-case clear-by” time).
- **Road-closure and priority probabilities.** The likelihood an incident escalates to a full road closure, and its operational priority.
- **Next-incident intensity.** A per-junction Hawkes rate λ (incidents/hour) that anticipates emerging hotspots.
- **Deployment plan.** Officer-team coordinates with head-counts, ranked barricade junctions, and diversion routes — each carrying a coverage/quality guarantee.
- **Economic translation.** Vehicle-minutes of congestion avoided per day, converted to person-hours and Rs. /day via a value-of-time model.

2.1 Headline Validated Metrics (Held-out Test Set)

Metric	Result
Survival concordance (verified-label incidents)	C-index 0.667 (GNN) vs. 0.663 (XGBoost)
Calibration (Integrated Brier Score)	0.0514 (CASCADE) vs. 0.0546 (XGBoost AFT)
Conformal coverage (target 90%)	90.7% empirical, 90.0% under online drift (ACI)
Next-incident concordance (Hawkes head)	C-index 0.645
RL dispatcher vs. greedy baseline	+ 20.3% congestion relieved
Plan vs. random placement (realised outcomes)	+ 360%
Digital-twin impact (illustrative)	~51% congestion avoided, ~Rs. 4.9M/day

Honesty note: only ~44% of incidents have a verified end-timestamp; the remainder use a proxy that injects label noise. We therefore report concordance on both the full and verified-label slices — on trustworthy labels the deep model genuinely wins.

3. Model Architecture and Working

CASCADE is a **contagion-aware, shared-trunk, multi-task system**. A single spatio-temporal encoder feeds several specialised heads; the outputs are calibrated and then consumed by a decision layer.

3.1 The Shared Trunk: City-State Encoder

The road network is modelled as a graph: 294 named junctions are nodes; edges fuse geographic k -nearest-neighbour proximity with corridor adjacency. The trunk combines:

- a **Graph Convolutional Network (GCN)** that blends each junction with its road-graph neighbours (the *spatial* mixer), and
- a **Gated Recurrent Unit (GRU)** that blends each incident with the recent history at its junction (the *temporal* mixer).

The result is a compact embedding H_t summarising “the situation at this junction now, given its neighbourhood and recent past.” We additionally implemented a **Mamba selective state-space** kernel as a drop-in temporal swap; empirically the GRU remained the better default for our short $L=8$ history windows, validating our conservative kernel choice.

3.2 Head 1 — Survival / Severity (DeepHit)

We discretise clearance time into K bins and predict a probability mass function plus closure and priority logits. The loss is censoring-aware: uncensored incidents contribute the log-likelihood of their bin, while censored incidents contribute the log-survival beyond their last-observed time.

$$\mathcal{L}_{\text{sv}} = \mathcal{L}_{\text{nll}} + \alpha \mathcal{L}_{\text{rank}}, \quad \mathcal{L}_{\text{nll}} = \begin{cases} -\log p_{k(i)} & \text{uncensored (event observed)} \\ -\log \sum_{k>k(i)} p_k & \text{censored (still active)} \end{cases} \quad (1)$$

The ranking term $\mathcal{L}_{\text{rank}}$ enforces correct ordering of cumulative incidence between pairs of incidents, directly improving concordance.

3.3 Head 2 — Self-Exciting Hawkes Intensity

Incident *genesis* is modelled as a temporal point process. The neural intensity λ is trained with the exponential point-process negative log-likelihood, where Δt is the time to the next incident at a junction and o its observation indicator:

$$\mathcal{L}_{\text{pp}} = -o \log \lambda + \lambda \Delta t, \quad \lambda(t, s \mid H) = \mu_s + \sum_{t_j < t} \alpha e^{-\beta(t-t_j)}. \quad (2)$$

This is precisely the contagion mechanism: a baseline rate μ_s plus an excitation term that decays after each recent nearby event. It is what allows CASCADE to forecast *the next* hotspot.

3.4 Multi-Task Fusion: MMoE + GradNorm $\rightarrow$ PCGrad

Naïvely summing head losses causes *negative transfer*. We route the trunk through a Multi-gate Mixture-of-Experts (MMoE), then resolve conflicting gradients on the *shared* trunk with gradient surgery: we GradNorm-normalise each task’s gradient, then apply PCGrad projection so the survival and point-process objectives stop fighting over the representation. Heads are balanced with learnable Kendall uncertainty weights:

$$\mathcal{L}_{\text{CASCADE}} = \sum_{\tau \in \{\text{pp}, \text{sv}, \text{cal}, \text{dfl}\}} \frac{1}{2\sigma_\tau^2} \mathcal{L}_\tau + \log(\prod_\tau \sigma_\tau). \quad (3)$$

3.5 Calibration: Conformal Prediction + Adaptive Conformal Inference

A raw “90%” is meaningless without a guarantee. Using a held-out calibration split disjoint from training, split-conformal prediction produces an offset q such that the upper bound $U(x) = \hat{q}_{1-\alpha}(x) + q$ achieves the target coverage. Because retraining and censoring break exchangeability, we deploy **Adaptive Conformal Inference** online:

$$\alpha_{t+1} = \alpha_t + \gamma (\alpha_{\text{target}} - \text{err}_t), \quad (4)$$

which automatically widens the bounds when coverage slips and tightens them when it over-covers — holding 90.0% coverage even under temporal drift.

3.6 Decision Layer: OR-Tools, RL Dispatcher, and SPO+

- **OR-Tools allocator.** A CP-SAT Maximal-Coverage model places officer teams to maximise “severity-minutes” covered within a response radius; a budgeted ILP selects barricade junctions by closure-probability $\times$ spillover. A predictive term folds in the Hawkes intensity so resources are *pre-positioned*.
- **GATE-PPO dispatcher.** A graph-encoder reinforcement-learning policy learns to dispatch the fleet over a simulated day, beating the greedy baseline by **+20.3%** by trading travel cost against congestion and anticipated arrivals.
- **SPO+ (decision-focused learning).** We implemented the Smart-Predict-then-Optimise objective; it confirmed that, for this budgeted top-*B* dispatch decision, the two-stage severity is already near-optimal — a rigorous, honest finding that *validates* our design.
- **Digital twin.** A deterministic queueing model (with an optional SUMO micro-corridor) quantifies vehicle-minutes saved and translates them into Rs. /day.
- **Closed loop.** De-censoring of resolved incidents, intensity residuals, and off-policy reward feed back into the system; the deployed plan beats random placement by **+360%** on realised outcomes.

4. Code and Technical Implementation

4.1 Tech Stack and Code Structure

The pipeline is fully reproducible from the raw CSV and is organised by layer:

- **Data & Graph:** `pandas`, `numpy`, `scikit-learn`, `networkx` — ingestion with survival labelling, junction-graph construction, leakage-safe causal “ripple” features, and a portable training bundle.
- **Models (GPU, Colab bridge):** `PyTorch` GCN+GRU trunk, DeepHit and Hawkes heads, MMoE/PCGrad fusion; a CPU build is used locally to smoke-test every notebook before GPU training.
- **Calibration:** custom conformal-survival and Adaptive Conformal Inference modules.
- **Optimisation:** Google OR-Tools (CP-SAT) allocator, `networkx` diversion routing.
- **Reinforcement Learning:** a `numpy` dispatch environment plus a `PyTorch` graph-encoder PPO policy, trainable and validated locally.
- **Baselines & Metrics:** `XGBoost` (AFT/Cox), `lifelines`, `scikit-survival` (concordance, Integrated Brier Score).

Every layer emits a versioned artifact (Parquet/NPZ/JSON), so stages are independently runnable and each figure is regenerable from the data.

4.2 Website Integration: The Command Center

The judge-facing prototype is an **enterprise “command-center” dashboard** built with **Streamlit**, GPU-accelerated **pydeck** (deck.gl) maps on a Mappls/CARTO basemap, and **Plotly** visualisations.

UI/UX. A dark, slate-and-amber design system organises the application into four progressive-disclosure tabs — *Command Center*, *Model Transparency*, *Economic Impact*, and *Deployment Worklist* — eliminating scrolling fatigue. The hero element is a 3D extruded congestion-density map over labelled Bengaluru, with a one-click **naive** ↔ **corrected** toggle that visually *is* the project’s core insight. Supporting elements include predictive insight cards, a **Commander’s Briefing** that translates the optimisation into plain English (explainable AI), animated semi-circle Hawkes risk gauges, a DeepHit survival visualiser, and a self-running “Live Network Pulse” that streams simulated time-of-day load, active incidents, and an incident ticker.

User interaction with the model. A sidebar of *System Parameters* lets an officer set the budget — number of officer teams, barricades, response radius, and the predictive-vs-reactive weight — and a *focus junction* for inspection. A time-of-day slider re-simulates network demand across the map.

Backend handling of predictive data. For speed and robustness the dashboard reads precomputed artifacts (calibrated bounds, allocation, diversions, economic report) rather than recomputing models live. The single interactive *re-optimisation* call invokes the OR-Tools allocator in an isolated subprocess (a deliberate engineering choice that sidesteps a platform-specific solver/runtime conflict) and re-renders the plan in seconds. A unified serving layer fuses all model outputs into one recommendation payload plus a grounded operations briefing. The same dashboard ships with a lightweight deploy requirements file and Dockerfile for one-URL hosting.

5. Anticipated Challenges and Future Scaling

- **Label quality.** Only $\sim 44\%$ of incidents carry a verified end-time; the proxy duration caps achievable concordance. A field data-capture improvement (mandatory closure timestamps) would lift every model simultaneously.
- **Conformal validity under the closed loop.** Continuous retraining and censoring break exchangeability; we mitigate with Adaptive Conformal Inference and conformalised survival, but large-scale online deployment will require ongoing re-drawing of disjoint calibration sets.
- **Live data and real-time geometry.** Moving from a static log to a streaming feed requires incremental graph updates and rate-limit-tolerant integration of external routing services (Mappls), kept strictly as a progressive enhancement with an offline fallback so the system never fails on an API outage.
- **RL robustness and the feedback loop.** Scaling the GATE-PPO dispatcher to a city-wide, multi-agent setting raises training-stability and reward-design challenges, and demands an explicit equity constraint to avoid amplifying the historical patrol-exposure bias.

- **Causal generalisation.** Distinguishing whether enforcement *reduces* congestion or merely *displaces* it (difference-in-differences, displacement analysis) is the next frontier for ensuring CASCADE targets interventions that actually stick.

CASCADE turns a passive incident log into a calibrated, contagion-aware decision system — not another map of the past, but a trustworthy plan for the next hour.